

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering

CSE 3102, Lab No. 10, Lab work

1.Create a database called **“LibraryDB”**. Write the corresponding query for this in your report.

2.Create a table titled **“Books”** with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
book_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
title	VARCHAR(100)	NOT NULL
author	VARCHAR(100)	NOT NULL
genre	VARCHAR(50)	NOT NULL
year_published	YEAR	NOT NULL

3.Create a table titled **“Members”** with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
member_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
member_name	VARCHAR(100)	NOT NULL
join_date	DATE	NOT NULL
email	VARCHAR(100)	NOT NULL

4.Create a table titled **“Borrowings”** with the following constraints. Write the corresponding query for this in your report

Field	Type	Constraints
borrowing_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
member_id	INT	NOT NULL
book_id	INT	NOT NULL
borrow_date	DATE	NOT NULL
return_date	DATE	NULLABLE

5. Populate the table **“Books”** with the following data. Write the corresponding insertion query for this in your report.

title	author	genre	year_published
The Alchemist	Paulo Coelho	Fiction	1988
Clean Code	Robert C. Martin	Programming	2008
The Pragmatic Programmer	Andy Hunt	Programming	1999
Goblet of Fire	J.K. Rowling	Fantasy	2000

6.Populate the table **“Members”** with the following data. Write the corresponding insertion query for this in your report.

member_name	join_date	email
Ayesha Rahman	2023-01-10	ayesha@gmail.com

Tanvir Ahmed	2023-02-20	tanvir@gmail.com
Nusrat Jahan	2023-03-05	nusrat@gmail.com

7. Populate the table “**Borrowings**” with the following data. Write the corresponding insertion query for this in your report.

member_id	book_id	borrow_date	return_date
1	1	2023-04-01	2023-04-10
2	2	2023-04-03	NULL
1	3	2023-04-05	NULL
3	4	2023-04-07	2023-04-20

After creating and populating the tables, write queries in your report to do the following tasks.

a) Query 1: Display all books.

Sample Output:

book_id	title	author	genre	year_published
1	The Alchemist	Paulo Coelho	Fiction	1988
...	...	...	...	...

b) Query 2: List all member names and emails.

Sample Output:

member_name	email
Ayesha Rahman	ayesha@gmail.com

Tanvir Ahmed	tanvir@gmail.com
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c) Query 3: Display all books borrowed but not yet returned.

Sample Output:

member_id	book_id	borrow_date
2	2	2023-04-03
1	3	2023-04-05

d) Query 4: Find books in the genre "Programming".

Sample Output:

title	genre
Clean Code	Programming
The Pragmatic Programmer	Programming

e) Query 5: Count the number of books borrowed by each member.

Sample Output:

member_name	total_borrowed
Ayesha Rahman	2
Tanvir Ahmed	1

f) Query 6: Show members who borrowed more than one book.

Sample Output:

member_name	total_books
Ayesha Rahman	2

g) Query 7: Find the average borrow duration (in days) for returned books.

Sample Output:

avg_days
11.5

h) Query 8: Show borrowing history of “Ayesha Rahman”.

Sample Output:

title	borrow_date
The Alchemist	2023-04-01
The Pragmatic Programmer	2023-04-05

i) Query 9: List the most borrowed book(s).

Sample Output:

title	borrow_count
The Pragmatic Programmer	1
Clean Code	1

j) Query 10: Show total books borrowed per genre.

Sample Output:

genre	total_borrowed
Fiction	1
Programming	2
Fantasy	1

Query Hints:

Query No	Concepts Used
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1–2	Simple SELECT
3–4	SELECT + WHERE
5	GROUP BY
6	HAVING
7	AVG + DATEDIFF()
8	JOIN + WHERE
9–10	GROUP BY + ORDER BY + COUNT

Solution:

1. Create the database

```
CREATE DATABASE LibraryDB;
```

2. Create Books table

```
CREATE TABLE Books ( book_id INT PRIMARY KEY AUTO_INCREMENT, title VARCHAR(100) NOT NULL, author VARCHAR(100) NOT NULL, genre VARCHAR(50) NOT NULL, year_published YEAR NOT NULL
```

```
);
```

3. Create Members table

```
CREATE TABLE Members ( member_id INT PRIMARY KEY AUTO_INCREMENT, member_name VARCHAR(100) NOT NULL, join_date DATE NOT NULL, email VARCHAR(100) NOT NULL
```

```
);
```

4. Create Borrowings table

```
CREATE TABLE Borrowings (
```

```
    borrowing_id INT PRIMARY KEY AUTO_INCREMENT,  
    member_id INT NOT NULL,    book_id INT NOT NULL,  
    borrow_date DATE NOT NULL,    return_date DATE,  
    FOREIGN KEY (member_id) REFERENCES Members(member_id),  
    FOREIGN KEY (book_id) REFERENCES Books(book_id)  
);
```

5. Insert into Books

```
INSERT INTO Books (title, author, genre, year_published) VALUES  
( 'The Alchemist', 'Paulo Coelho', 'Fiction', 1988),  
( 'Clean Code', 'Robert C. Martin', 'Programming', 2008),  
( 'The Pragmatic Programmer', 'Andy Hunt', 'Programming', 1999),  
( 'Goblet of Fire', 'J.K. Rowling', 'Fantasy', 2000);
```

6. Insert into Members

```
INSERT INTO Members (member_name, join_date, email) VALUES  
( 'Ayesha Rahman', '2023-01-10', 'ayesha@gmail.com'),  
( 'Tanvir Ahmed', '2023-02-20', 'tanvir@gmail.com'),  
( 'Nusrat Jahan', '2023-03-05', 'nusrat@gmail.com');
```

7. Insert into Borrowings

```
INSERT INTO Borrowings (member_id, book_id, borrow_date, return_date) VALUES  
(1, 1, '2023-04-01', '2023-04-10'),  
(2, 2, '2023-04-03', NULL),  
(1, 3, '2023-04-05', NULL),  
(3, 4, '2023-04-07', '2023-04-20');
```

Query a: Display all books

```
SELECT * FROM Books;
```

Query b: List all member names and emails

```
SELECT member_name, email FROM Members;
```

Query c: Books borrowed but not yet returned

```
SELECT member_id, book_id, borrow_date FROM Borrowings  
WHERE return_date IS NULL;
```

Query d: Books in genre 'Programming'

```
SELECT title, genre FROM Books  
WHERE genre = 'Programming';
```

Query e: Number of books borrowed by each member

```
SELECT M.member_name, COUNT(*) AS total_borrowed  
FROM Borrowings B  
JOIN Members M ON B.member_id = M.member_id  
GROUP BY B.member_id;
```

Query f: Members who borrowed more than one book

```
SELECT M.member_name, COUNT(*) AS total_books  
FROM Borrowings B  
JOIN Members M ON B.member_id = M.member_id  
GROUP BY B.member_id
```

HAVING total_books > 1;

Query g: Average borrow duration (in days)

```
SELECT AVG(DATEDIFF(return_date, borrow_date)) AS avg_days  
FROM Borrowings  
WHERE return_date IS NOT NULL;
```

Query h: Borrowing history of Ayesha Rahman

```
SELECT B.title, BR.borrow_date  
FROM Borrowings BR  
JOIN Books B ON BR.book_id = B.book_id  
JOIN Members M ON BR.member_id = M.member_id  
WHERE M.member_name = 'Ayesha Rahman';
```

Query i: Most borrowed book(s)

```
SELECT B.title, COUNT(*) AS borrow_count  
FROM Borrowings BR  
JOIN Books B ON BR.book_id = B.book_id  
GROUP BY BR.book_id  
ORDER BY borrow_count DESC  
LIMIT 1;
```

Query j: Total books borrowed per genre

```
SELECT B.genre, COUNT(*) AS total_borrowed  
FROM Borrowings BR  
JOIN Books B ON BR.book_id = B.book_id
```

GROUP BY B.genre;

THE END